

Auteur: Siebe Mees
 Studierichting: Informatica
 Oefeningenreeks 2B nr. 5.3

$$\begin{aligned}
 & \forall x > 0 : \text{Bgtan}(x) + \text{Bgtan}\left(\frac{1}{x}\right) \\
 & \left\{ \begin{array}{l} y = \text{Bgtan}(x) \iff \tan(y) = x \text{ en } y \in \left]0, \frac{\pi}{2}\right[\\ z = \text{Bgtan}\left(\frac{1}{x}\right) \iff \tan(z) = \frac{1}{x} \text{ en } z \in \left]0, \frac{\pi}{2}\right[\end{array} \right. \\
 & \tan z = \frac{1}{\tan y} = \cot y = \tan\left(\frac{\pi}{2} - y\right) \rightarrow z = \frac{\pi}{2} - y \\
 & \Rightarrow \text{Bgtan}(x) + \text{Bgtan}\left(\frac{1}{x}\right) \\
 & = y + z \\
 & = y + \left(\frac{\pi}{2} - y\right) \\
 & = \frac{\pi}{2}
 \end{aligned}$$

References